CO_2 Adsorption on Modified Mg–Al-Layered Double Hydroxides

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ABSTRACT: Because of increases in the burning of fossil fuels, there is an increased emission of pollutant gases (e.g. CO_2) into the atmosphere. An alternative to minimize these emissions is to capture these gases using porous materials, such as layered double hydroxides (LDHs). In this study, LDHs were synthesized by co-precipitation of ion salts of Mg^{2+} and Al^{3+}. After the synthesis, the materials were treated with a template to expand the layers. The samples were then characterized by X-ray diffraction, thermogravimetric analysis/derivative thermogravimetry, adsorption/desorption of nitrogen, scanning electron microscopy/energy-dispersive X-ray spectroscopy and transmission electron microscopy. A CO_2 adsorption study was then performed to determine the adsorption capacity of the material in accordance with the contact time between the gas and the adsorbent. The adsorption capacities of LDHs and LDHs-P123 for CO_2 were 0.72 and 1.36 mmol/g, respectively, showing satisfactory results post-treatment.

1. INTRODUCTION

Greenhouse gases, naturally present in the atmosphere, form a layer that retains heat radiated by the Earth. Among the various greenhouse gases, carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O) are notable (Cotton and Pielke 1995). Carbon dioxide is a component of the atmosphere that is essential for plant life and some marine organisms. However, increased emissions of CO_2 into the atmosphere are considered the main reason for greenhouse effect, supposedly responsible for 60% of the temperature increase in the atmosphere. This process of increase in temperature is termed *global warming* (Yamakasi 2003). Among the various source of CO_2 emissions, approximately 30% are generated by power plants that use fossil fuels, which are subsequently the major contributors to global warming (Benson and Orr 2008).

One potentially economical and less problematic alternative to carbon capture and sequestration technology that has been widely investigated is adsorption using solid physical and chemical adsorbents (Diban *et al*. 2013; Lv *et al*. 2012; Zhao *et al*. 2014). At present, the technological challenge is finding an adsorbent that has a high adsorption capacity and good selectivity while also having good regeneration capacity (desorption), thereby enabling its use in an industrial scale. In order for a material to be considered ideal and have good efficiency, it should adsorb large amounts of CO_2 under mild conditions of temperature and pressure while also enabling the total desorption of CO_2, thereby leaving their adsorption sites regenerated for a new step in the process. The hydrotalcites or layered double hydroxides (LDHs) show high stability and have basic surface (Bellotto *et al*. 1996; Yong *et al*. 2001). These properties are of great importance in the process of adsorption, making them a promising material for CO_2 adsorption.

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Recently, LDH have been extensively synthesized and studied by several authors (Kagunya *et al*. 1996; Tsujimura *et al*. 2007; Wang *et al*. 2012b). In the current study, we synthesized LDH according to the procedure described by Reichile *et al*. (1986). After the synthesis, the material was modified by treatment with Pluronic P123 [(EO)20(PO)70(EO)20] assisted by hydrothermal treatment.

2. MATERIALS AND METHODS

2.1. Synthesis

The LDHs were synthesized by co-precipitation based on the procedure described by Reichile *et al*. (1986). The molar ratio of Mg to Al used in the synthesis was 2.0. Initially, a solution of $Mg(NO_3)_2{\cdot}6H_2O$ and $Al(NO_3)_3{\cdot}9H_2O$ was prepared for the synthesis of LDHs in distilled water (solution A). This solution was added dropwise to a basic solution B (NaOH and Na_2CO_3 in distilled water) by dripping of reagents with constant stirring at room temperature. The thus-obtained gel was aged at 60 °C for 24 hours, under constant stirring. The resulting suspension was filtered and the solid obtained was washed with distilled water to maintain the pH between 8 and 9. The material was dried at 105 °C for 2 hours. After drying, the precursors were pulverized and subjected to calcinations at 550 °C.

A pre-determined quantity of LDHs that had been synthesized and calcined was treated with a triblock co-polymer—Pluronic P123 (EO20PO70EO20). In the first stage, the co-polymer and hydrochloric acid (HCl) were mixed with distilled water (H_2O). The mixture was stirred continuously until the co-polymer P123 dissolves completely. This solution was then added to the sample kept under magnetic stirring at a temperature of approximately 60 °C for 24 hours. After the 24-hour stirring, the gel formed was transferred into a stainless steel autoclave where it was kept for 48 hours at a temperature of approximately 100 °C. After heating for 48 hours, the material was cooled down to room temperature and then washed with a solution of hydrochloric acid (2%) in ethanol to aid in the removal of the organic P123 triblock co-polymer. After the washing step, the material was vacuum filtered using a Buchner funnel (Zhao *et al*. 1998). The pH of the reaction was adjusted to a value of 10 by adding a solution of NaOH (4 M; Wang *et al*. 2010). After the hydrothermal ageing step, the sample was filtered and dried at 100 °C in an oven. The co-polymer surfactant was then removed by calcination at 550 °C.

2.2. Characterization

The powder X-ray diffraction patterns of the samples were obtained using Cu-K_α radiation with a diffractometer (Rigaku model Miniflex II). The data were collected in the 2θ range of 5–70°. The nitrogen adsorption/desorption experiments were performed at 77 K using NOVA-1200e. The material synthesized was pre-treated by heating it to 120 °C for 3 hours. The adsorption and desorption isotherms were obtained in the partial pressure (p/p_0) range of 0.01–0.95. The specific surface area of samples was determined by the BET method and the volume and pore diameter were calculated by the Barrett–Joyner–Halenda method. Thermogravimetric and derivative thermogravimetry analyses were performed using a thermogravimetric analyzer (TGA-50; Shimadzu). The samples were heated from 25 °C to approximately 900 °C, with a heating rate of 10 °C/minute and under a dynamic atmosphere of nitrogen (flow rate, 25 ml/minute). Scanning electron micrographs were obtained using a JEOL SM 840 microscope, operating at 15 kV. Transmission electron microscopy (TEM) images were obtained using a Philips CM 200 microscope, operating at 10 kV. The samples were placed on an aluminium drum and metallized with a gold film using the JFC-1100E ION SPUTTER. The energy-dispersive X-ray spectroscopy measurements were captured using an Edax CM200ST probe with an SiLi detector. The material was dispersed in 2-propanol and dropped onto a copper grid.

2.3. Adsorption Experiments

The efficiency of the adsorbents after the heat treatment was evaluated through the process of CO_2 adsorption. A thermogravimetric balance (DSC/TGA SDT Q600; TA Instruments) was used for this purpose. The materials were subjected to a pre-treatment process *in situ*, which involved heating at 140 °C under a flow of nitrogen at 50 ml/minute for 30 minutes. This process is necessary for the removal of water. After this step, the adsorption of CO_2 begins. The adsorption temperatures were fixed at 30, 40, 50 and 60 °C and the samples were saturated with a flow of 50 ml/minute for 90 minutes.

3. RESULTS AND DISCUSSION

3.1. Characterization Studies: X-Ray Diffraction, Electron Microscopy Scanning and Scanning Electron Microscopy

Figure 1 shows the diffraction patterns for LDH and LDH-P123 in the (003), (006) and (009) planes along with their peak values (degrees), which indicate the formation of a crystal structure layered with rhombohedral symmetry (3R; Othman *et al*. 2006). It can also be seen that only a lower fraction of the periclase phase (MgO) was formed in the plane [(200), represented by asterisks] relative to the diffraction peak at 2θ = 43.03° (JCPDS 003-0998).

When treated with P123, the sample showed the same peak characteristics as that of the material before treatment, but with reduced intensity, providing evidence that the structure has remained the same even after treatment with P123 [Figure 1(b)]. Three planes of diffraction peaks (2θ = 37.01°, 43.00° and 62.44°) are assigned to MgO (asterisks), which showed well-defined peaks, indicating that the mixed oxide formed is cubic MgO doped with an amorphous species (Al_2O_3; Wang *et al*. 2010). The appearance of impurity (i.e. chloride, which is represented by

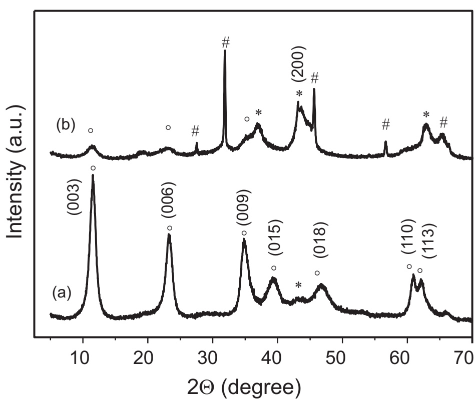

Figure 1. X-ray diffraction pattern: (a) LDH and (b) LDH-P123.

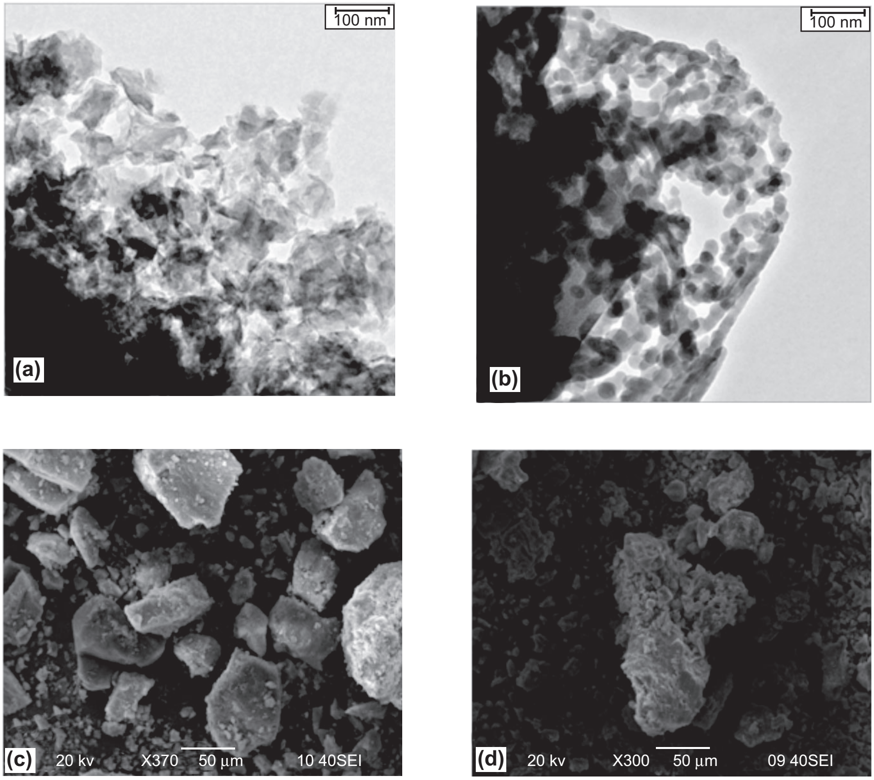

Figure 2. TEM images: (a) LDH and (b) LDH-P123. Scanning electron microscopy images: (c) LDH and (d) LDH-P123.

octothorpes in the figure) was also observed. After calcination at 550 °C, the material still showed the (003) diffraction peak, which can be explained as follows: Mg–Al and Zn–Al LDHs have the regeneration feature, called *memory effect*, because these materials lose the ability to regenerate only when subjected to heat treatments above 873 K, as a result of complete decomposition of their hydroxyl group, subsequently producing stable crystalline phases of the oxides. This regeneration can occur by simple exposure of the calcined LDH to atmospheric CO_2, with intercalation of carbonate anions (Delorme *et al*. 2007; Puttaswamy *et al*. 1997; Rives 2014; Vaccari 1998; Son *et al*. 2010). This effect is only observed when LDHs are treated to certain temperatures, beyond which the thermal decomposition becomes irreversible due to the formation of stable phases such as spinel and MIIM2IIIO4 MIIO. In the case of hydrotalcites, the spinel phase is formed close to 1000 °C (Velu *et al*. 1999; Daza *et al*. 2012). These temperatures depend on the MII and MIII metals and the anions present in the LDH structure (Bellotto *et al*. 1996).

Figure 2 shows the micrographs of LDHs and LDHs-P123. Figures 2(c and d) show particles with irregular surfaces, structures not well defined and particle aggregates with different sizes. Note that the flat shape of the particle is consistent with the lamellar structure. It can be seen that the blades of LDHs are stacked as perpendicular sheets about the surface (Wang *et al*. 2010).

In Figures 2(a and b) it can be seen that the LDHs aggregate to form sheets that are not stacked. The material appears as grains with irregular distribution and as sheets with curved shapes. The disordered structure of the oxides is clearly evident from the TEM images (Benito *et al*. 2009).

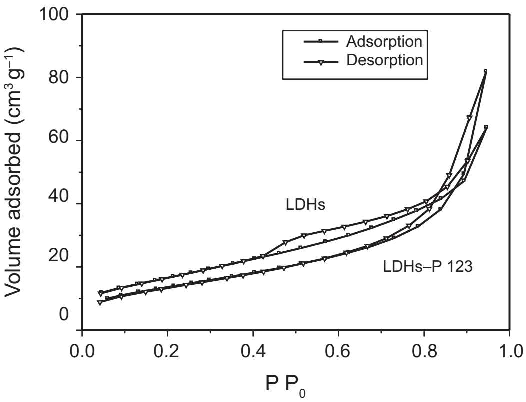

Figure 3. Adsorption/desorption isotherms of LDHs and LDHs-P123.

TABLE 1. Elementary Components and Texture Properties of Samples

	Chemical composition (%)			Mg/Al	S_{BET}	Dp	Nf
	Mg	Al	Cl	ratio	(m^2/g)	(nm)	(mmol/g)
LDHs	16.9	10.4	–	1.75	50.1	1.53	0.72
LDHs-P123	15.52	12.4	4.81	1.28	61.1	1.8	1.36

Nf = maximum adsorption capacity.

When materials are subjected to heat treatment, they lose their crystallinity, and thus there is a breakdown in their structure, which may be one explanation for the decrease in the intensity of the peak at d(003). In contrast to LDHs, the LDHs-P123 particles are rounded and have better organization and shapes with a diameter of approximately 7.8 nm.

3.2. Textural and Chemical Analysis

The adsorption/desorption isotherms (Type III) of LDHs both in the absence and in the presence of P123 are shown in Figure 3. The hysteresis type H3 is presented, characterized when the interaction between adsorbate and the adsorbent is low. The isotherms showing an increase in surface area, volume and pore diameter for the LDHs-P123 samples are also shown. The heat treatment was thus effective in improving the dimensions of the sample material.

The elemental composition of the samples are presented in Table 1. The minimum formula for LDHs and LDHs-P123 is Mg0, 70Al0, 40 (Mg/Al = 1.75) and Mg0, 64Al0, 50 (Mg/Al = 1.28), respectively. The numerical values in this formula refer to the number of moles of each element present in these materials. When Mg/Al = 2 was used in the synthesis, part of the Mg was not involved in the structure formation. The appearance of Cl in the material composition corroborates with the results found in X-ray diffraction, which indicates chloride contamination in the material.

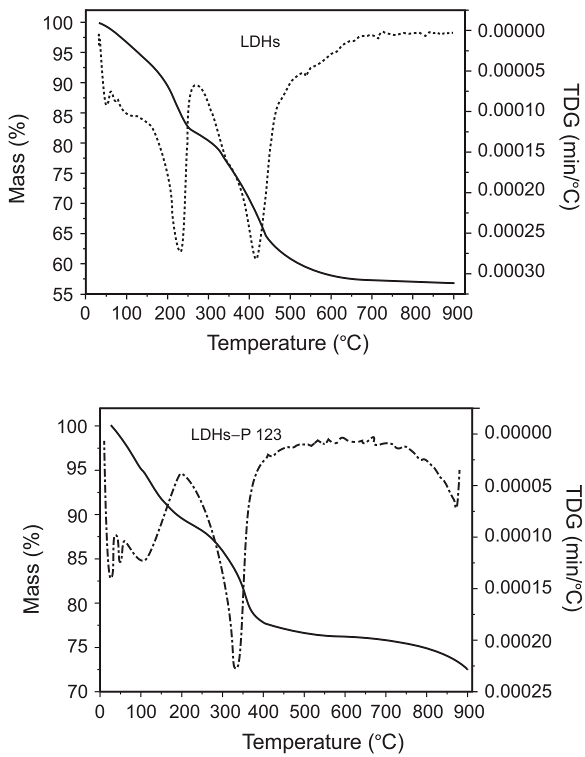

Figure 4. Thermogravimetric curves: (a) LDHs and (b) LDHs-P123.

3.3. Thermogravimetry Analysis

Figures 4(a and b) show the thermogravimetric curves of LDHs and LDHs-P123. The two mass losses are presented in Figure 4(a): the first mass loss (i.e. 17.21%) occurs from room temperature to about 250 °C, and is caused by the removal of water and part of the hydroxyl and carbonate groups. The second mass loss of 24.75% occurs from 250 to 630 °C, which is related to the removal of the remaining portion of hydroxyl group. After these losses, the structure turns into an oxide, thus destroying the lamellar phases. However, for the treated sample, three mass losses are evident. The first occurring at room temperature and up to 180 °C, equivalent to approximately 6% mass loss relative to the removal of physisorption water and structural changes. The second loss (approximately 11%) between 200 and 450 °C is relative to the removal of carbonate possibly regenerated when the material was exposed to air and part of the hydroxyl group. The third loss (2.8%) from 750 to 900 °C is probably due to the remaining hydroxyl group forming an oxide of magnesium and aluminium. This delay in the loss of part of the hydroxyl group may have been caused by the treatment with the template.

3.4. Studies of Adsorption of CO_2

The CO_2 interacted with the LDHs rapidly and the maximum uptake was observed within 40 minutes (Figures 5 and 6). Afterwards, the interactions slowed down and approached equilibrium in nearly 90 minutes under the given set of experimental conditions. In Figure 5, it can be seen

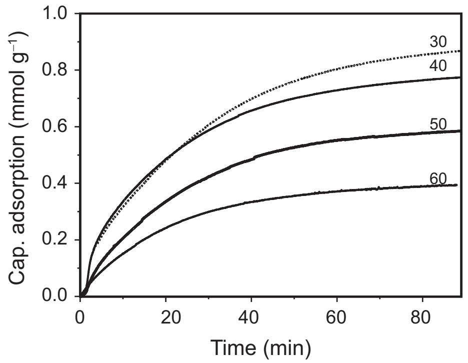

Figure 5. Adsorption of CO_2 on LDHs at different temperatures.

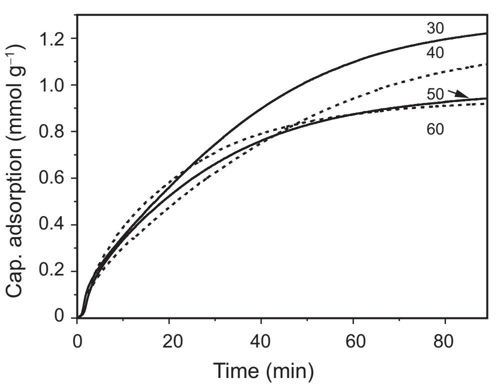

Figure 6. Adsorption of CO_2 on LDHs-P123 at different temperatures.

that among the various temperatures studied, better results for the adsorption of CO_2 were obtained at 30 °C. Similar result was obtained for LDHs-P123 as well (Figure 6). Thus, this temperature was selected for the calculation of adsorption capacity. The adsorption capacity was calculated using the following formula:

$$n = \Delta m/MM \text{ and } N_f = n/m_{(ads)} \tag{1}$$

where n is the number of moles of carbon dioxide adsorbed; Δm is the mass variation (%) in sample weighing; MM is the molar mass of carbon dioxide; N_f is the adsorption capacity of adsorbent and m (ads) is the initial mass of adsorbent or without adsorbed gas (pure).

Figure 7 shows the CO_2 adsorption capacity. The maximum uptake was 0.72 and 1.36 mmol/g for LDHs and LDHs-P123, respectively. The LDHs-P123 samples showed similar adsorption behaviour to that of LDH samples, along with reduction in the amount of gas following adsorption as well as a decrease in the contact time between the gas and the sample with the increase in temperature. The better adsorption capacity for the template-treated LDH samples can be attributed to the textural properties of the material. Treatment with the template increased both the surface area and the pore diameter of the original material (Table 1).

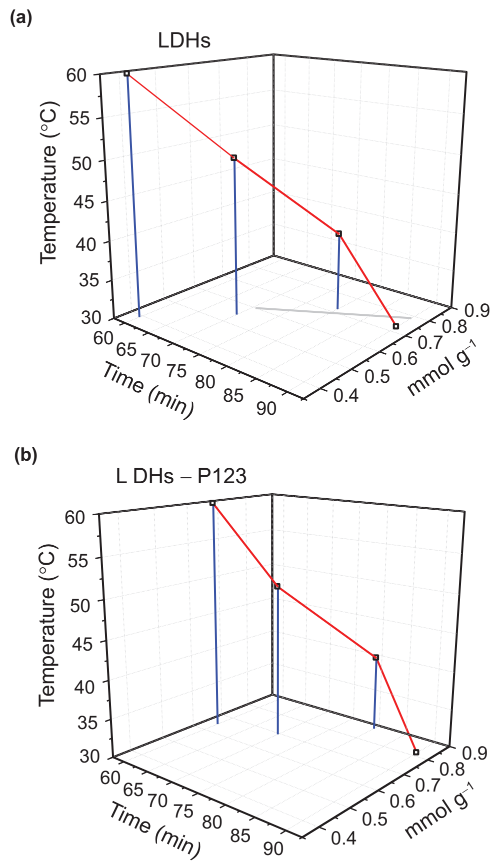

Figure 7. CO_2 adsorption capacity for (a) LDHs and (b) LDHs-P123.

According to Wang *et al*. (2010), both the physical and chemical properties of LDHs influence their CO_2 capture capacity. Wang *et al*. (2012b) recently published a systematic investigation on the synthesis of Mg–Al hydrotalcites at different pH values. The CO_2 adsorption capacity was in the range of 0.1–0.83 mmol/g. The CO_2 adsorption capacity of 1.36 mmol/g of the LDHs-P123 samples reported in this study was the highest thus far reported for the Mg–Al LDHs samples among various studies in the literature (Gao *et al*. 2013; Wang *et al*. 2012b) under the experimental conditions studied. The amount of carbon dioxide adsorbed on LDH-P123 was approximately 1.3 mmol/g in the temperature range tested, which is within the range of 0.57–2.05 mmol/g reported in the literature for CO_2 adsorption on amine-modified LDHs (Wang *et al*. 2012a) and amine-modified mesoporous silica (Hiyoshi *et al*. 2005; Huang *et al*. 2003; Zelenak *et al*. 2008).

4. CONCLUSIONS

Following the synthesis, characterization studies were performed. The results showed that the LDHs synthesized by the co-precipitation method with constant pH had good crystallinity, even after the treatment with co-polymer P123, corroborating with the results reported in other studies.

The sample that showed greater textural areas had better CO_2 adsorption capacity, as expected. At higher temperatures the adsorption capacity was decreased, which consequently, decreased the contact time between the gas and the sample.

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